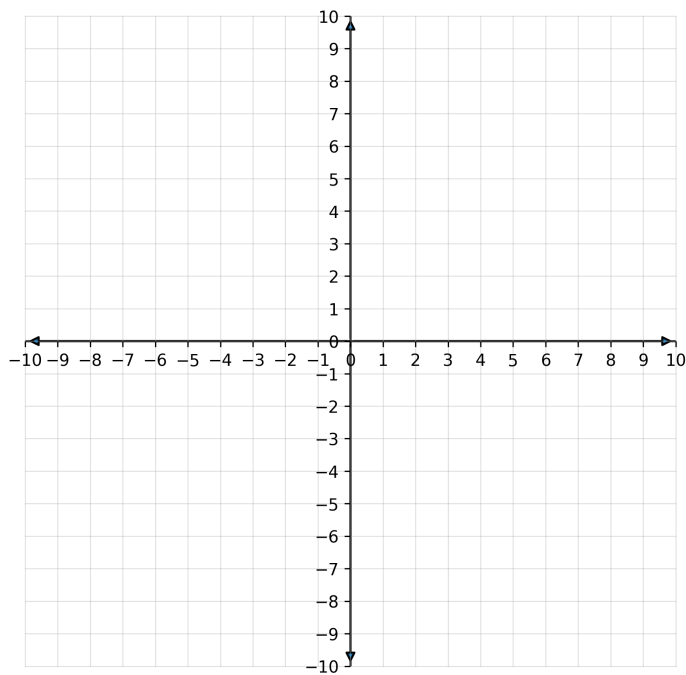


Algebra 2 - Unit 1 Test B

Name: _____

Graph each quadratic and find the requested features.

1. $f(x) = -(x - 3)^2 + 5$



vertex

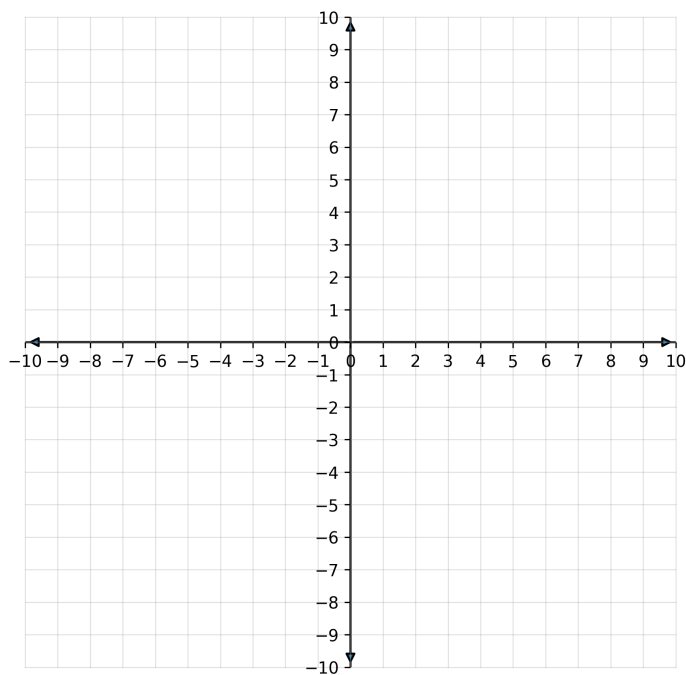
axis of symmetry

y-intercept

range

max or min

2. $f(x) = \frac{1}{2}x^2 + 2x - 5$



vertex _____

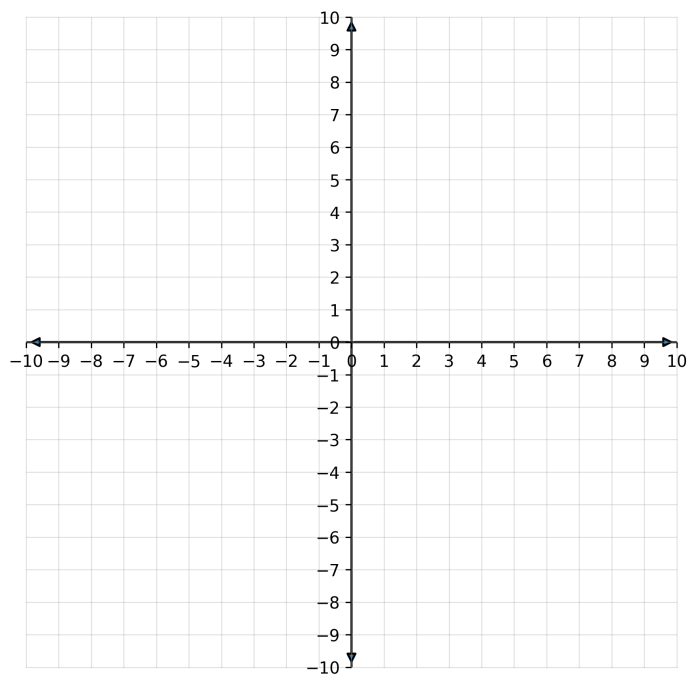
axis of symmetry _____

y-intercept _____

range _____

max or min _____

3. $f(x) = \frac{1}{4}(x + 4)(x - 4)$



vertex

axis of symmetry

y-intercept

range

max or min

x-intercepts

1. Roger threw a ball off of a cliff. The height of the ball (ft) over time (s) is well modeled by $h(t) = -\frac{1}{10}t^2 + 2t + 10$.

- a. How long does it take the ball to reach its maximum height?

- b. How high does the ball go?

- c. How high is the cliff?